

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 29-05-2018

Day: Tuesday

Time: 10:00 to 10:30 pm

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : MPT- May 2018
Semester: 1 st
Section : I
Course Code : PT861
Course Name: Musculoskeletal Imaging in Physiotherapy

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(10 × 1 = 10)

1. Radiograph has been referred to may terms, except:
 - a. Mariograph
 - b. Plain-film radiograph
 - c. Hard copy
 - d. Soft copy
2. Sun ray appearance is seen in _____
 - a. Osteoclastoma
 - b. Fibrous dysplasia
 - c. Osteosarcoma
 - d. Chondrosarcoma
3. Investigation of choice in traumatic paraplegia is _____
 - a. MRI
 - b. CT Scan
 - c. Myelography
 - d. Spine X – ray

4. Following are the traditional units used to express a quality of ionizing radiation except:
 - a. Roentgen (R)
 - b. Radiation absorption dose (RAD)
 - c. Curies (Ci)
 - d. Bequerel (Bq)
5. X-rays are modified _____
 - a. Protons
 - b. Electrons
 - c. Neutrons
 - d. Positrons
6. The recommended projection of osseous pelvis is _____
 - a. AP projection
 - b. PA projection
 - c. Lateral projection
 - d. Oblique projection
7. Computed tomography X-ray techniques allow the test component to be:
 - a. Viewed in various cross-sectional slices
 - b. Viewed from different angles
 - c. Analyzed for chemical composition
 - d. Wide range of wavelengths
8. Ionizing radiation can be used in industrial radiography because the health hazards:
 - a. Have been eliminated with controls and procedures
 - b. Are minimized through controls and procedures
 - c. Are worth the risk
 - d. Are being ignored
9. The number of carpal bones seen in a radiograph of an infant is _____
 - a. 0
 - b. 5
 - c. 3
 - d. 2
10. In multi-slice CT scan, the thickness of dual slice is ____ slice per rotation.
 - a. 2.5 mm
 - b. 5 mm
 - c. 7.5 mm
 - d. 10 mm

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

- | | True | False |
|---|--------------------------|--------------------------|
| 11. Lighter resultant image represents greater tissue density | ☐ | ☐ |
| 12. Barium sulfate is used in upper & lower GIT as contrast study. | ☐ | ☐ |
| 13. Radiolucent is used to identify areas that have decreased radiodensity. | ☐ | ☐ |
| 14. Ultrasound can only image the cortical outline of bone. | ☐ | ☐ |
| 15. Average effective dose for whole body CT scan is 15mSv | ☐ | ☐ |

(P.T.O.)

Match the following**(5 × 1 = 5)**

16.	Subcutaneous fat along with muscle sheaths and around viscera	A. Gray/White
17.	Produces one cross-sectional image	B. Transmitted intensity
18.	No. of x-ray photons falling on an object	C. Bone
19.	Number of photons passing through	D. Axial CT
20.	Contrast medium: white outline	E. Gray black
		F. Volume CT
		G. Incident intensity

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.P.T. Examination

May 2018

PT861 MUSCULOSKELETAL IMAGING IN PHYSIOTHERAPY

Date: 29/05/2018

Time: 10:30 am – 1 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. MRI sequences
- b. Types of image Receptors
- c. Ottawa rules for ankle
- d. Radiological features of osteoporosis
- e. What are the limitations of ultrasound?
- f. Fluoroscopy
- g. Hounsfield scale

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Cobb method of measurement for scoliosis
- b. Why Physiotherapist has to study Imaging?
- c. X-ray of knee osteoarthritis
- d. Role of MRI in evaluation of soft tissue injuries in knee.
- e. Production of X-ray

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. Discuss the ABCs APPROACH for viewing x-ray with suitable examples.
- b. Discuss in details MRI versus CT scan
- c. Discuss the radiological approach for a 40-year-old man presents with low back ache.